Preprocessing refers to the transformation of raw images into a format that can be efficiently used by a machine learning model. It ensures uniformity, numerical stability, and compatibility with the model input.

During training, preprocessing performs two roles:

1. Standardization (resize, normalize, color conversion)
2. Augmentation (random transformations)

Augmentation introduces variability such as flips, rotations, and brightness changes, helping the model learn invariant features and reducing overfitting.

Inference preprocessing is strictly deterministic. The same resizing, normalization, and color handling used during training are applied, but no random transformations are introduced. This ensures that predictions remain stable and reproducible.

Preprocessing Before Model Training (Humanised Explanation)

Before training a machine learning model, the raw images collected from the real world cannot be used directly. These images usually come from different sources, devices, and environments, which means they vary in size, lighting, orientation, and quality. Preprocessing is applied at this stage to make the data suitable for learning.

The first goal of preprocessing during training is to bring all images into a common format. This includes resizing images to a fixed size, normalizing pixel values so that the model can process them efficiently, and converting color spaces if required. These steps ensure that the model receives consistent and numerically stable input.

In addition to standardization, training preprocessing often includes data augmentation. Augmentation means creating slightly modified versions of the same image using operations such as flipping, rotating, or adjusting brightness. This helps the model see multiple variations of the same data and prevents it from memorizing the training images.

Because the model is still learning at this stage, some level of randomness is allowed in preprocessing. Each time an image is passed to the model, it may look slightly different due to augmentation. This makes the model more robust and improves its ability to generalize to unseen data.

Overall, preprocessing before model training focuses on helping the model learn better and generalize well, rather than producing exact or repeatable outputs.

Preprocessing Before Model Inference (Humanised Explanation)

Once a model has been trained, it is used to make predictions on new, unseen images. These images may come from real users, cameras, or test datasets. Before passing them to the model, preprocessing is still required, but the approach here is different from training.

The main purpose of preprocessing during inference is to ensure that the input image matches the format the model was trained on. This includes resizing the image to the same dimensions, normalizing pixel values using the same scale, and applying the same color space conversion. These steps help the model interpret the input correctly.

Unlike training, no data augmentation is applied during inference. Augmentation introduces randomness, such as flipping or rotating images, which can change the appearance of the input. During prediction, the model should give stable and repeatable results, so the preprocessing must be consistent every time the same image is used.

Inference preprocessing is therefore deterministic, meaning the same input image will always result in the same processed output. This consistency is essential for reliable predictions, performance evaluation, and real-world deployment.

In summary, preprocessing before inference focuses on maintaining consistency and reliability, ensuring that the model receives inputs in the exact format it expects.